ENSCI 99

A Practical Guide to Environmental Choices

Air Pollution I

Exam 2: April 8th

Lectures 10-19 (30 MC questions)

Objectives

- The Structure of Earth's Atmosphere
- Weather, Climate, and Atmospheric Conditions
- Air pollution
  - Categories: Primary and Secondary
  - Major Air pollutants and Health Impacts
- What is being done to control air pollution?
  - Legislation
  - Technology
  - Better Choices

NYC – Canadian wildfires

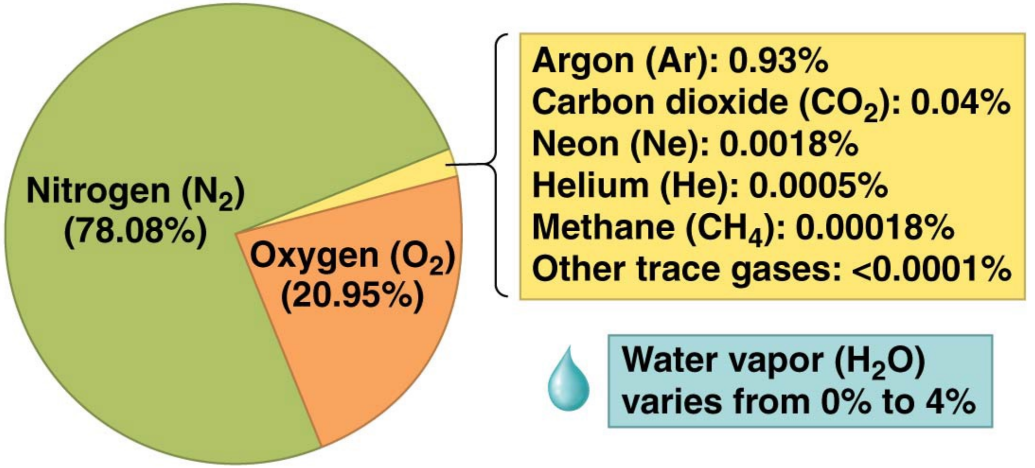
https://www.the-express.com/news/us-news/102855/orange-smog-canadian-wildfires-new-york-city

What are some health concerns related to polluted air?

What is the atmosphere?

A thin layer of <u>gases</u> around Earth

- Shield
- Provides oxygen
- Absorbs radiation and
  moderates climate

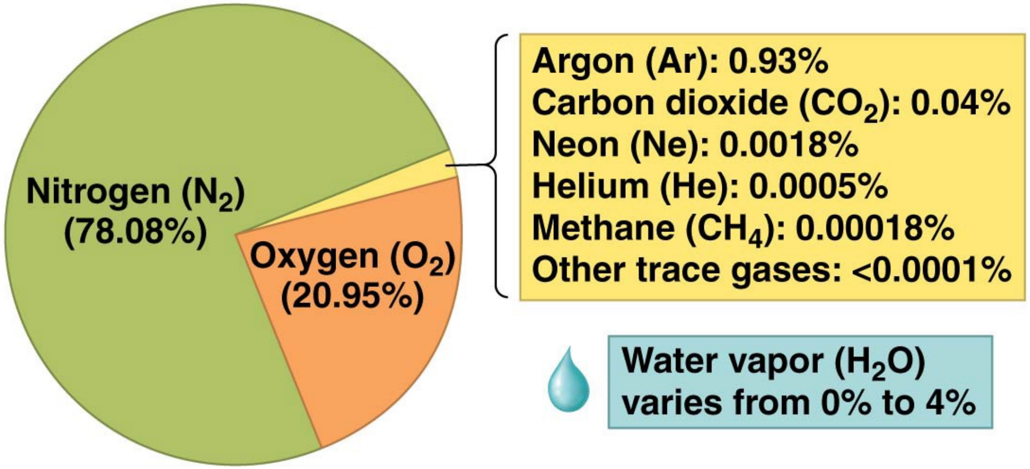


- Transports and recycles water and nutrients

78% N_2, 21% O_2, 1 % argon and trace amounts of other gases.

What is the atmosphere?

- Water vapor
- Trace amounts of
  - Argon
  - CO2
  - Neon
  - Helium
  - Methane
  - Trace gases

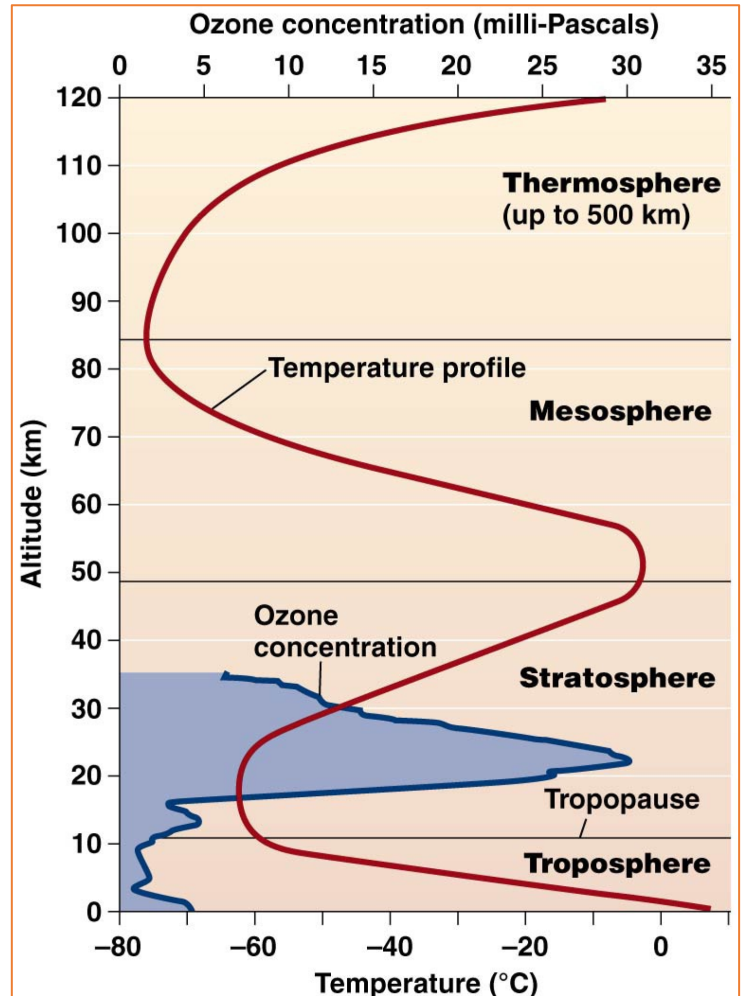


Human activity is now changing the amount of some gases
CO_2, methane (CH_4), ozone (O_3)

How many layers are there in the Earth's atmosphere?

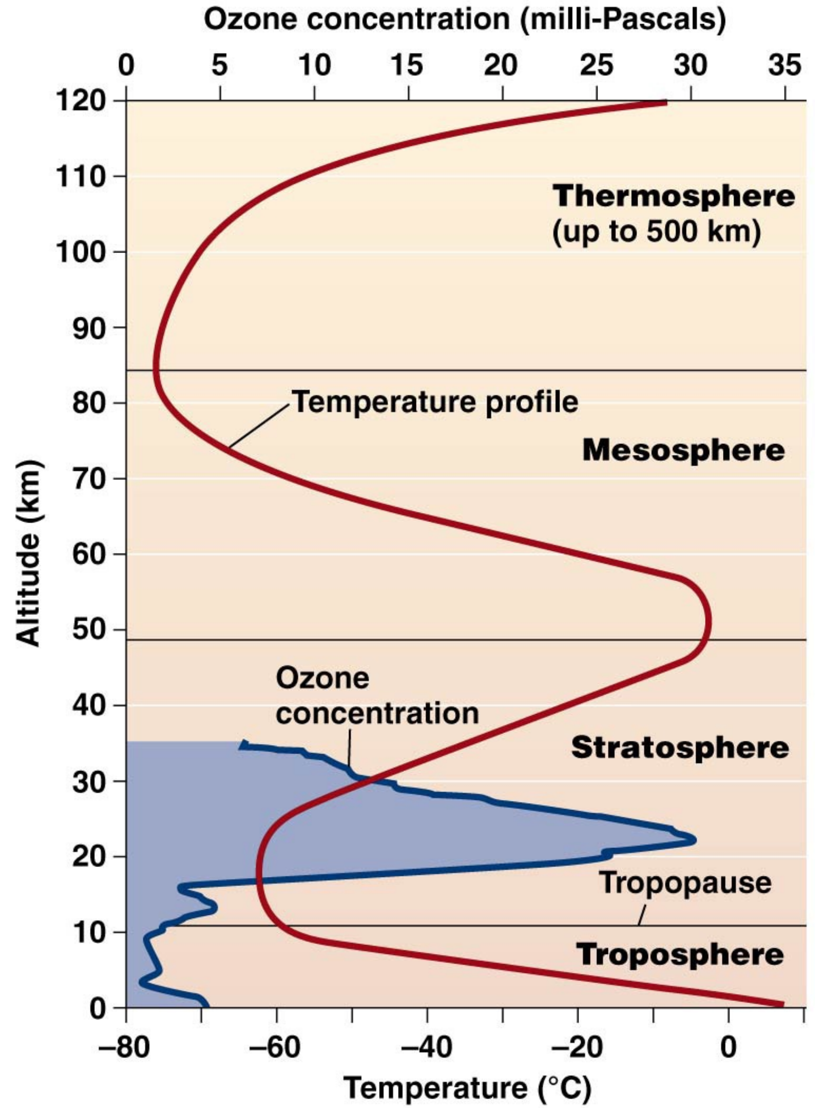
The atmosphere can be divided into four main layers.

- **Troposphere**
  - Bottommost layer (11 km [7 miles])
  - Most of the mass
  - Responsible for Earth's weather
  - **The air gets colder with altitude**



How many layers are there in the Earth's atmosphere?

- **Stratosphere**
  - 11–50 km (7–31 mi) above sea level
  - Drier and less dense, with **little vertical mixing**
  - Gets warmer with altitude
  - Many jet aircrafts fly here
  - **Ozone layer**
    - Blocks UV radiation



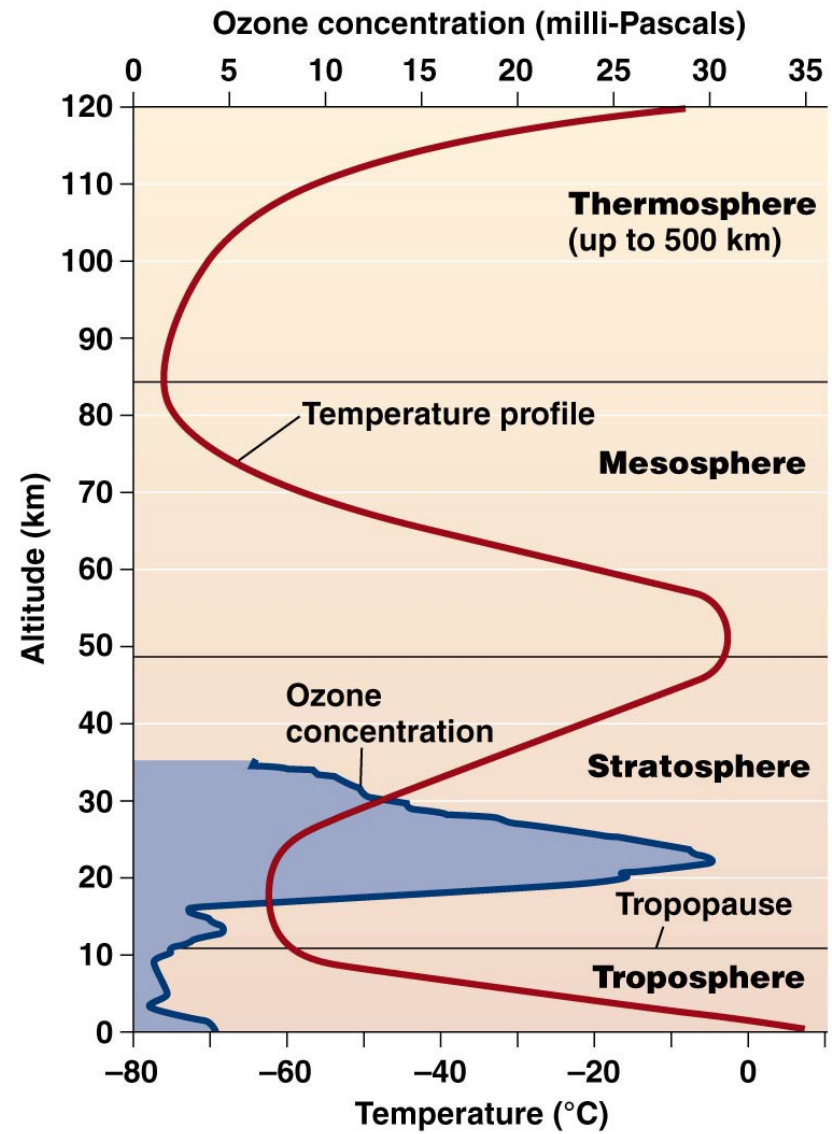
How many layers are there in the Earth's atmosphere?

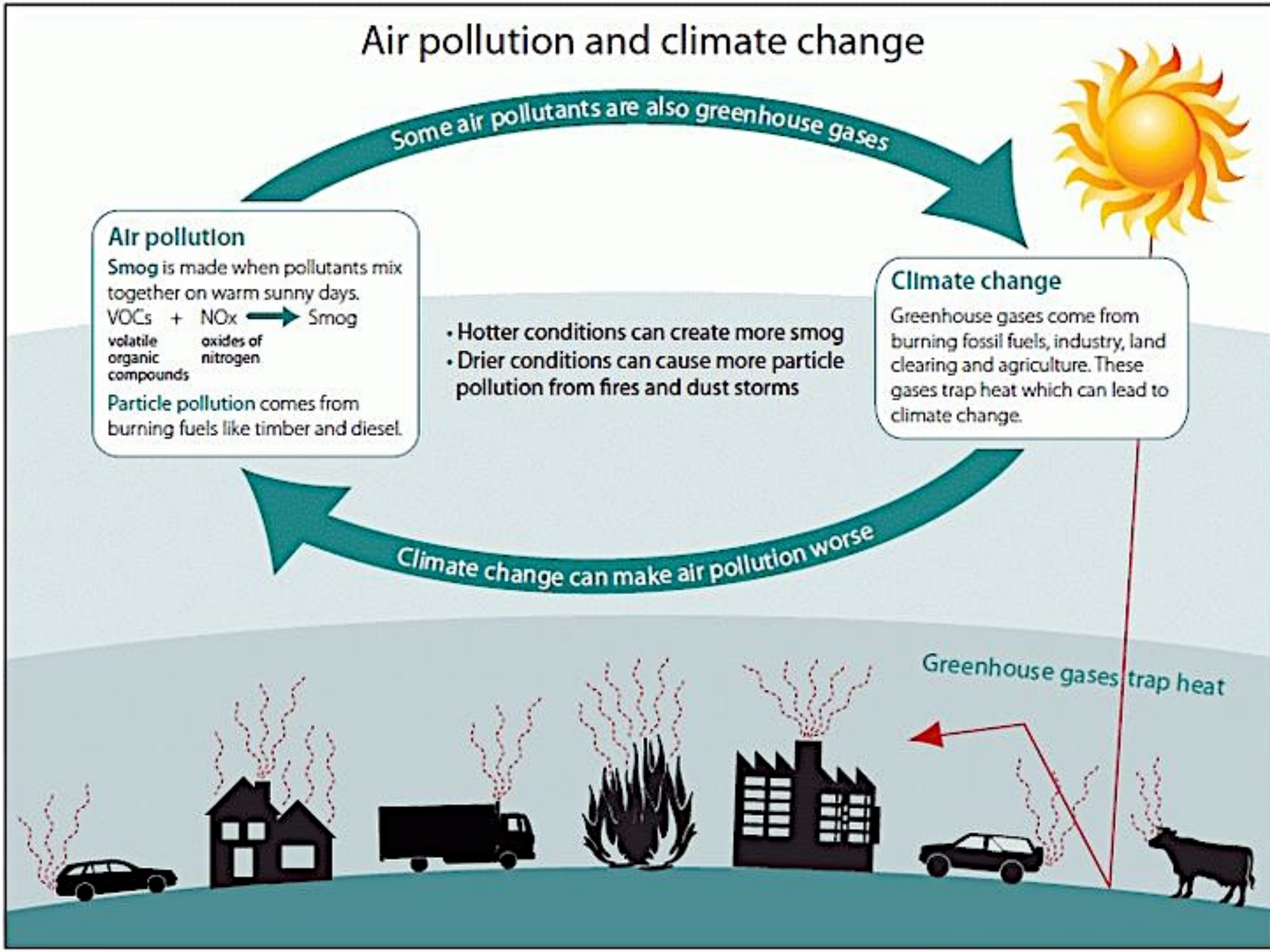
Mesosphere

- Low air pressure,
- Gets colder with altitude

Thermosphere

- Top layer
- The aurora mostly occur in the thermosphere



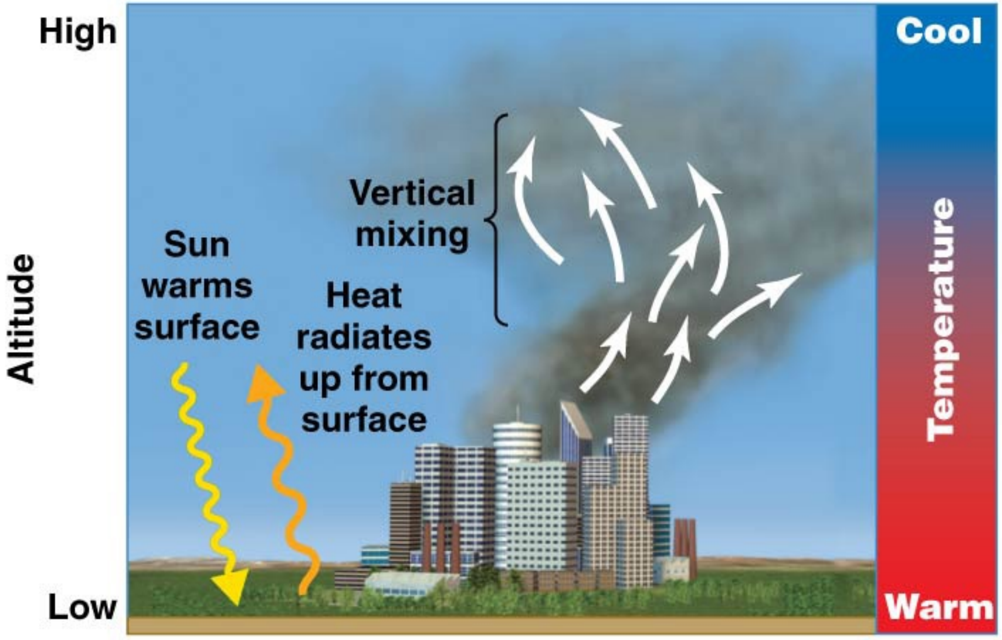


Adapted from State of the Environment NSW 2000

What is a thermal inversion?

Normal conditions

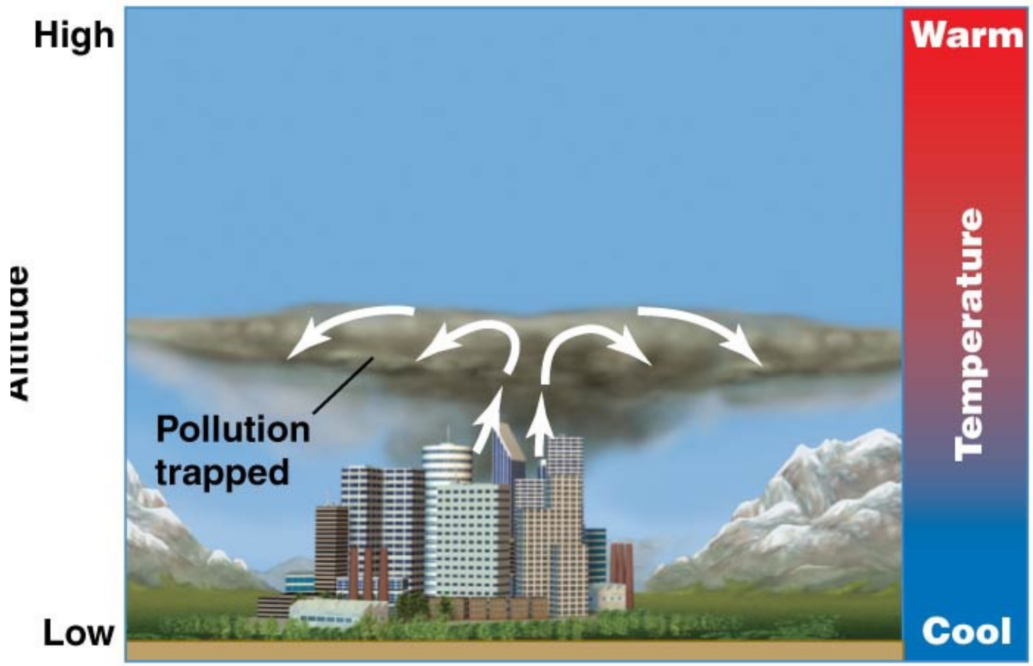
- Air become cooler with altitude
- Allows vertical mixing
- Dispersal of pollutants



(a) Normal conditions

Inversions affect air quality

- A layer of cool air forms beneath warm air
  - The band of air where temperature rises with altitude
  - Dense, cool air at the bottom of the layer resists mixing

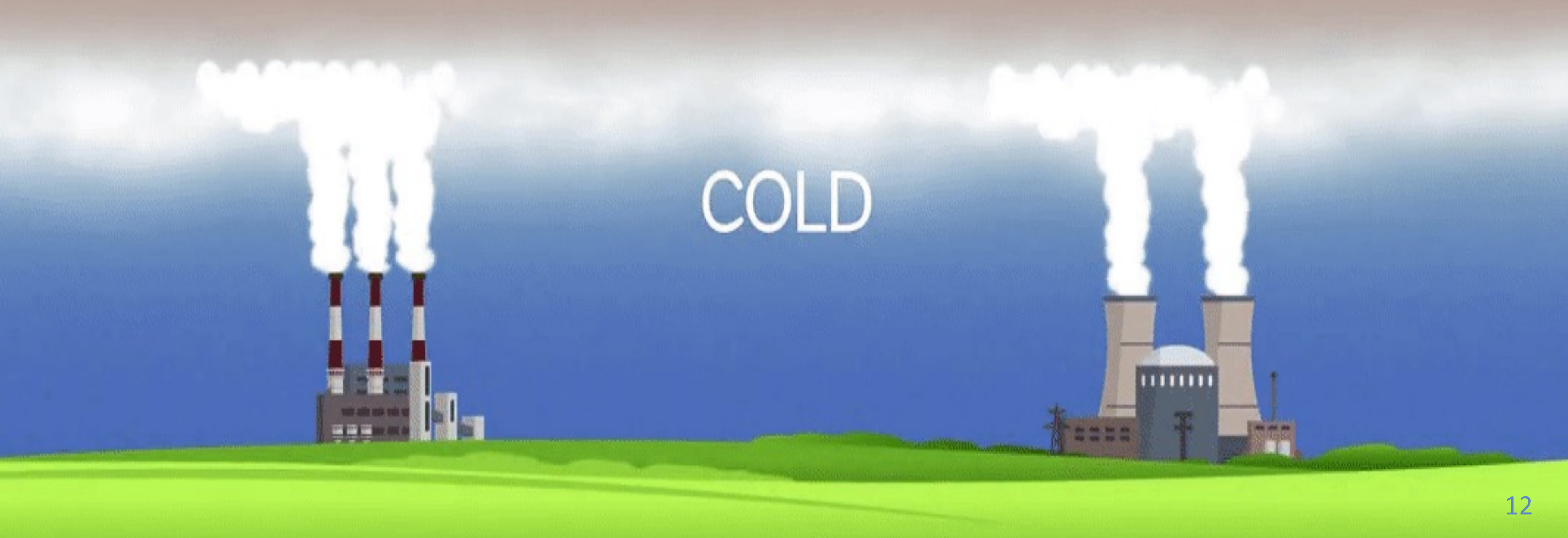


b) Thermal inversion

During a thermal inversion, a layer of cool air forms over a layer of warm air and traps air pollutants near the surface, leading to unsafe air quality.

Met Office

- In thermal inversions, the normal temperature patterns reverse.
  - The dense cold air doesn't rise.
  - Pollution accumulates.



1948 – Donora smog

- Thermal inversion
  - Trapped pollutants over the town
- Five-day episode - 1000's became ill, 20 people died in a week
- zinc smelting plant,
- steel mills' furnaces,
- sulfuric acid plant

Build up of CO_2, NO_2, SO_2, F_2

http://www.eoearth.org/view/article/151779/

What is air pollution?

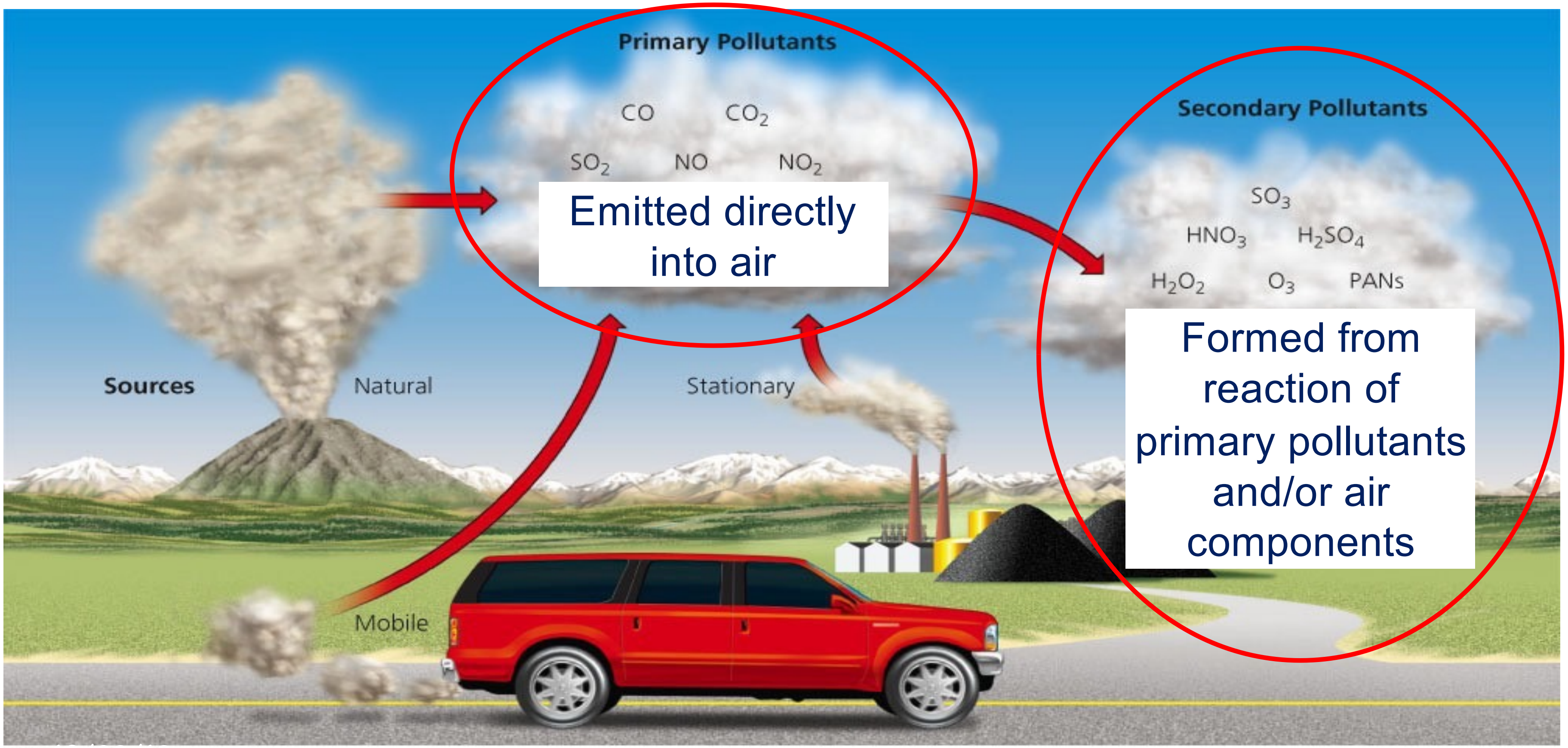
Substances added to the atmosphere by **natural** events or **human** activities in high enough concentrations to be harmful.

http://greenliving.lovetoknow.com/Types_of_Pollution

What are the **two main categories** of air pollution?

- <u>Primary pollutants</u> - products of natural events (like fires and volcanic eruptions) and human activities added directly to the air
- <u>Secondary pollutants</u> - formed by interaction of primary pollutants with each other or with normal components of the air

Types of Pollutants



Common types of air pollution

Carbon monoxide (CO)

- odorless, colorless, poisonous gas
- created by incomplete combustion (motor vehicles, industrial processes, burning biomass)
- combines with hemoglobin in blood & may create problems for infants, the elderly, & those with heart or respiratory diseases

The greatest sources of CO to outdoor air are cars, trucks and other vehicles or machinery that burn fossil fuels.

Holland tunnel air vent

https://www.epa.gov/co-pollution

Sulfur oxides

Mainly sulfur dioxide,(SO_2)
Produced largely from combustion of coal & oil (esp. coal)

- Responsible for acid rain problem
  - can react with gases in atmosphere to form sulfuric acid (**secondary air pollutant**)

Sulfur oxides

- Impairment of respiratory function
- Aggravation of existing respiratory disease (especially bronchitis)
- Decrease in the ability of the lungs to clear foreign particles
- Asthmatics, individuals with cardiovascular disease, elderly and children

Oxides of nitrogen

NO (nitric oxide) & NO_2 (nitrogen dioxide)

- Emitted directly by motor vehicles & industrial activities (burning fossil fuels)

- Responsible for acid rain problem
    - Can react with other gases in atmosphere to from nitric acid (HNO_3) (**secondary air pollutant**)

Volatile organic compounds (VOCs)

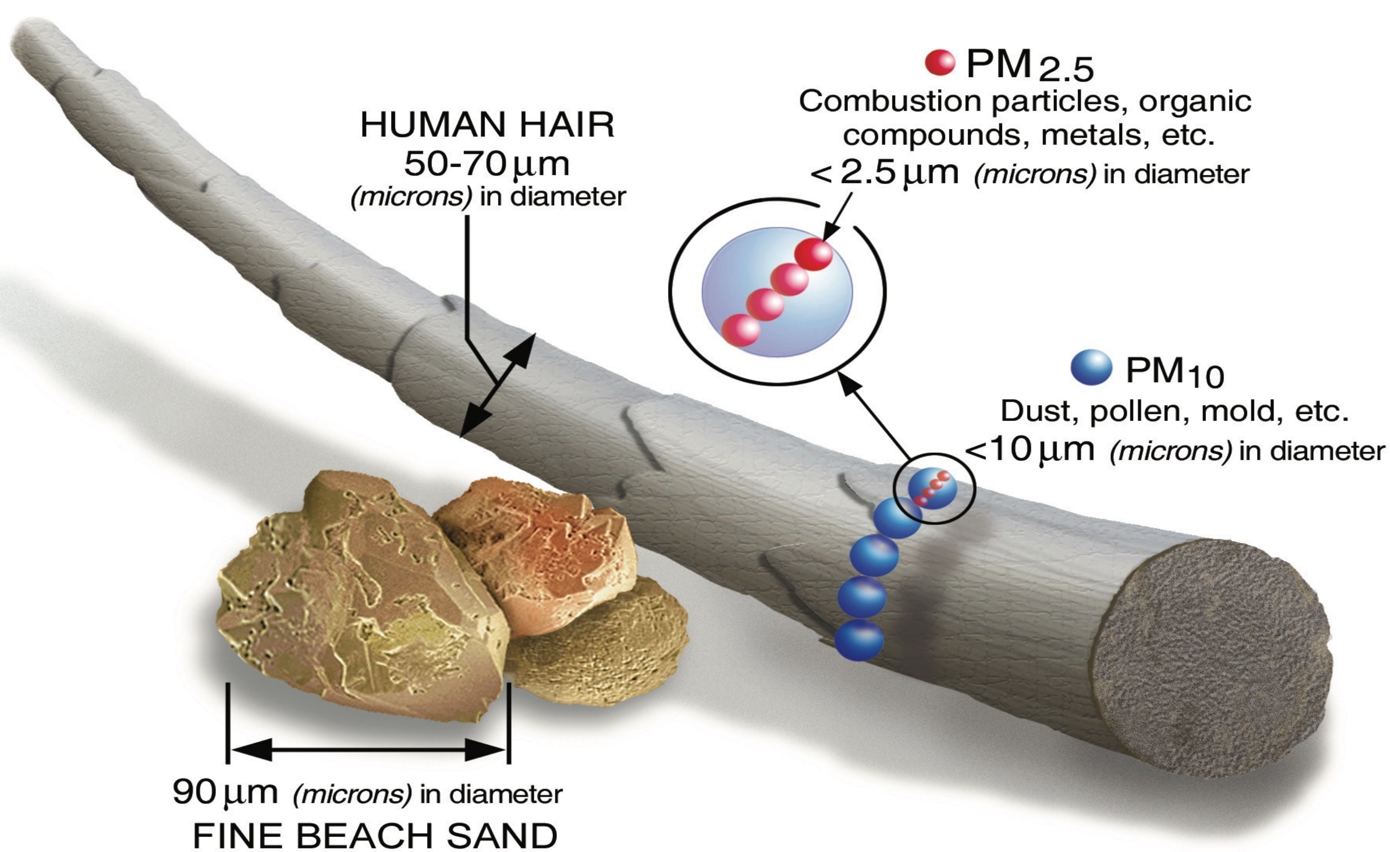
- Methane, benzene, propane, & chlorofluorocarbons (CFC's)
- **Sources include**: motor vehicles, industry, & various household products
- Eye and respiratory tract irritation, headaches, dizziness, visual disorders, and memory impairment
- VOCs are consistently higher indoors than outdoors.

Suspended particulate matter

Particulate Matter (dust, ash, salt particles)
Suspended solid or liquid particles

- Sources include incomplete combustion (smoke), power plants, iron/steel mills, land clearing, highway construction, mining
- Act as respiratory irritants; some are known carcinogens (e.g., asbestos)
- Can aggravate heart/respiratory diseases

Particulate matter 2.5, 10



Major PM2.5 pollution from **traffic** is not evenly distributed throughout the city

https://a816-dohbesp.nyc.gov/IndicatorPublic/traffic/index.html

Toxic compounds

Trace amounts of at least 600 toxic substances (such as **mercury** and **lead**) produced by human activities

- Sources of mercury are from burning coal and waste (such as medical wastes)
- Serious health effects on humans and wildlife

Ground level ozone (O_3)

Interaction between NO_x and VOCs in the presence of sunlight

- Summertime, urban areas, largely industrialized communities
- Can cause respiratory problems
- EPA provides maps showing levels of ozone pollution https://www.airnow.gov/

How is acid rain produced?

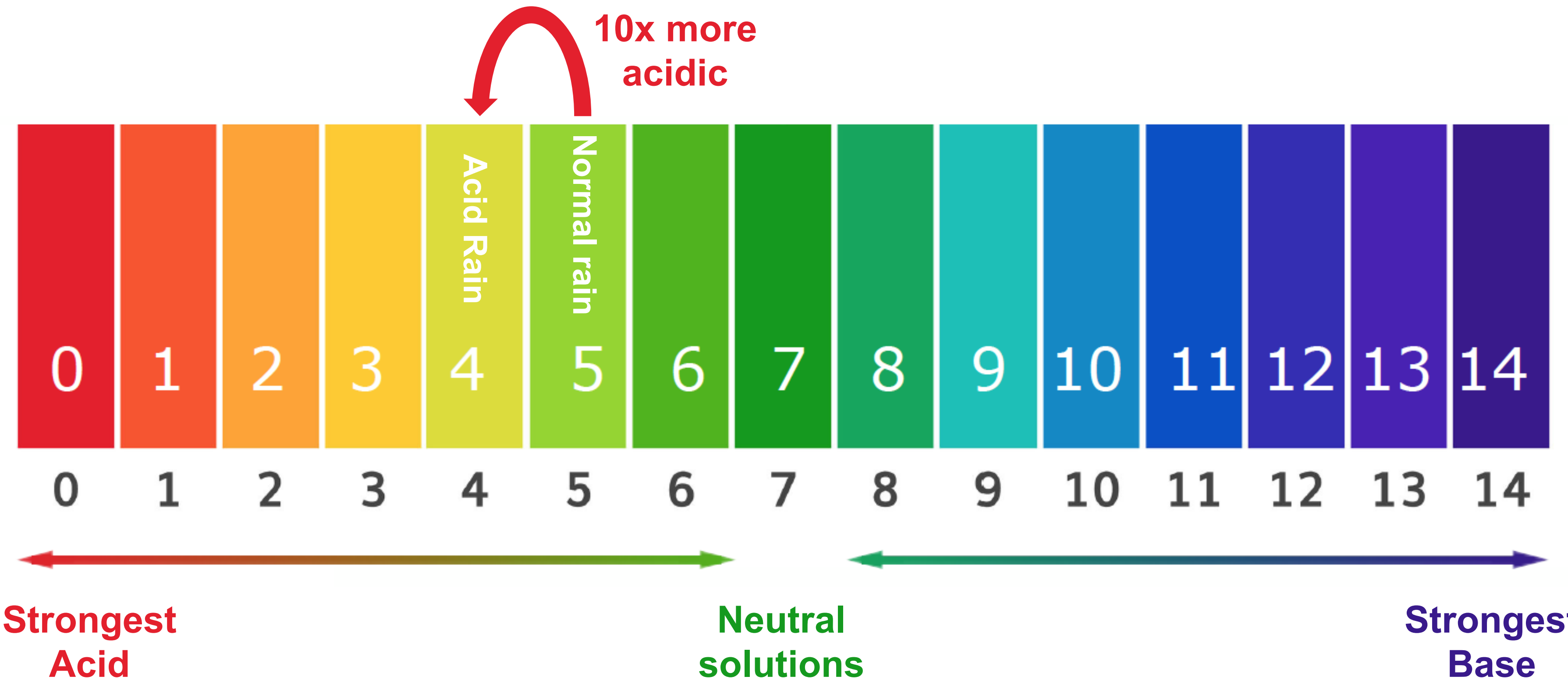
Sulfur dioxide (SO_2) and oxides of nitrogen (NO_x) can form acidic particles when they react with water:

$SO_2 + H_2O \rightarrow H_2SO_4$ (<u>sulfuric acid</u>)
$NO_x + H_2O \rightarrow HNO_3$ (<u>nitric acid</u>)

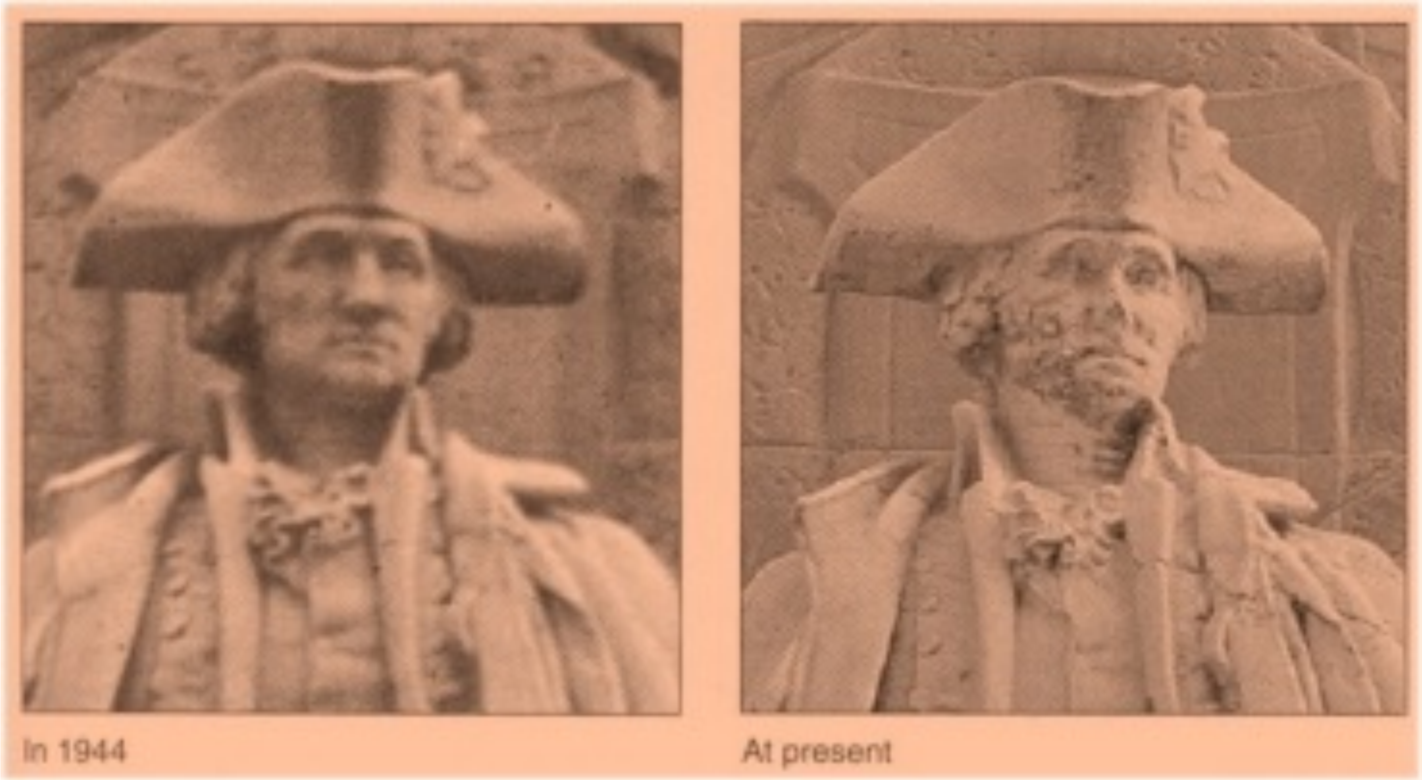
Impacts

1. Harmful to lakes, streams, forests
2. Leaching of nutrients such as calcium and potassium ions from topsoil.
3. Damaging to buildings, sculptures and statues

- **Acid precipitation** has a pH lower than normal rain.
  - Normal rain has a pH of about 5.6 due to carbonic acid.
  - Acid rain has a pH of 4.2-4.4 due to sulfuric and nitric acid.



Effects of acid rain



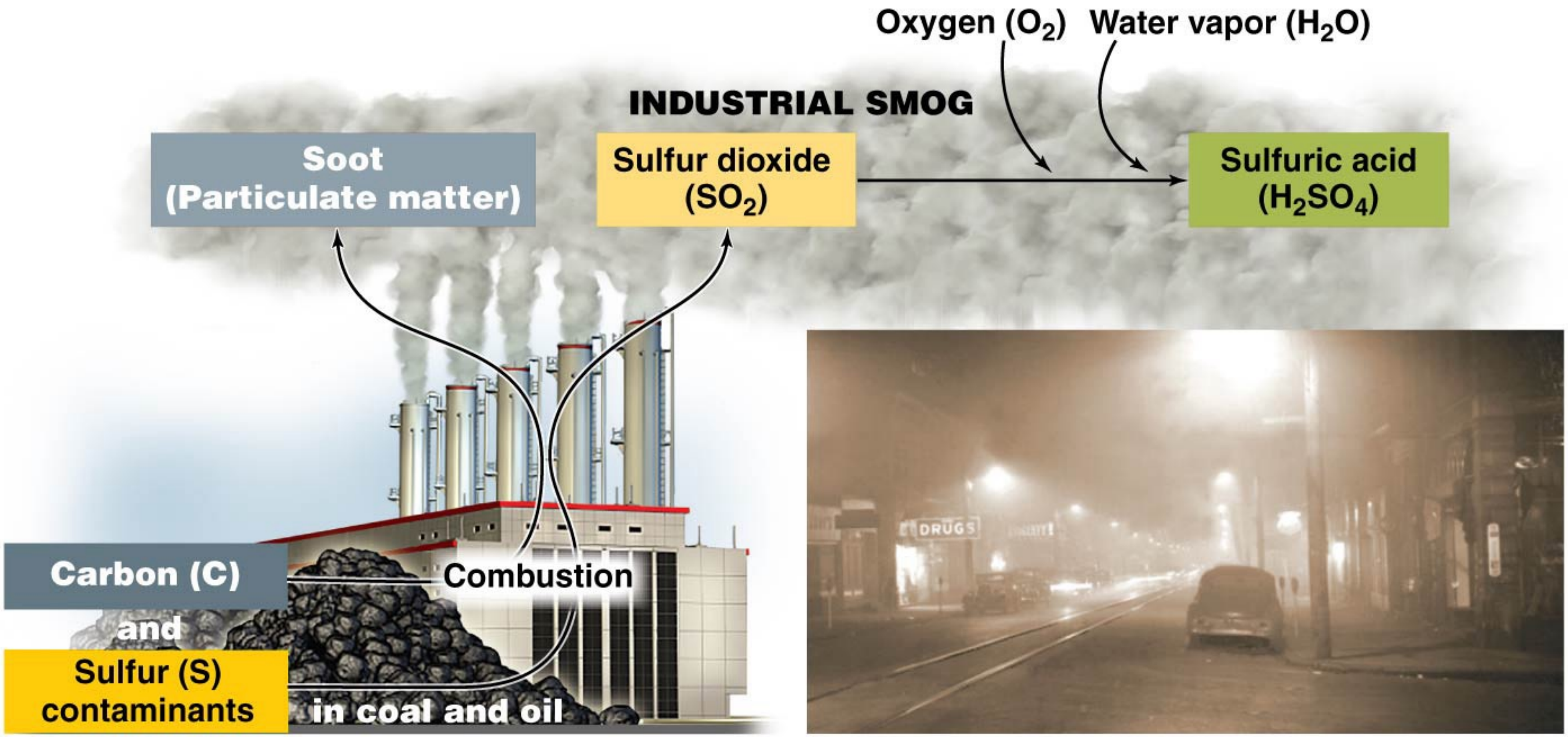
1944 **2002**

George Washington, New York

Acid rain is estimated to cause $5 billion in damages annually in a 17-state region by corroding buildings and other structures.

http://www.nytimes.com/1985/07/18/us/draft-study-puts-acid-rain-damage-at-5-billion-for-17-states.html

Industrial smog results from coal and oil combustion



(a) Formation of industrial smog

(b) Donora, Pennsylvania, at midday in its 1948 smog event

- Mix of smoke, sulfur oxides & sulfuric acid

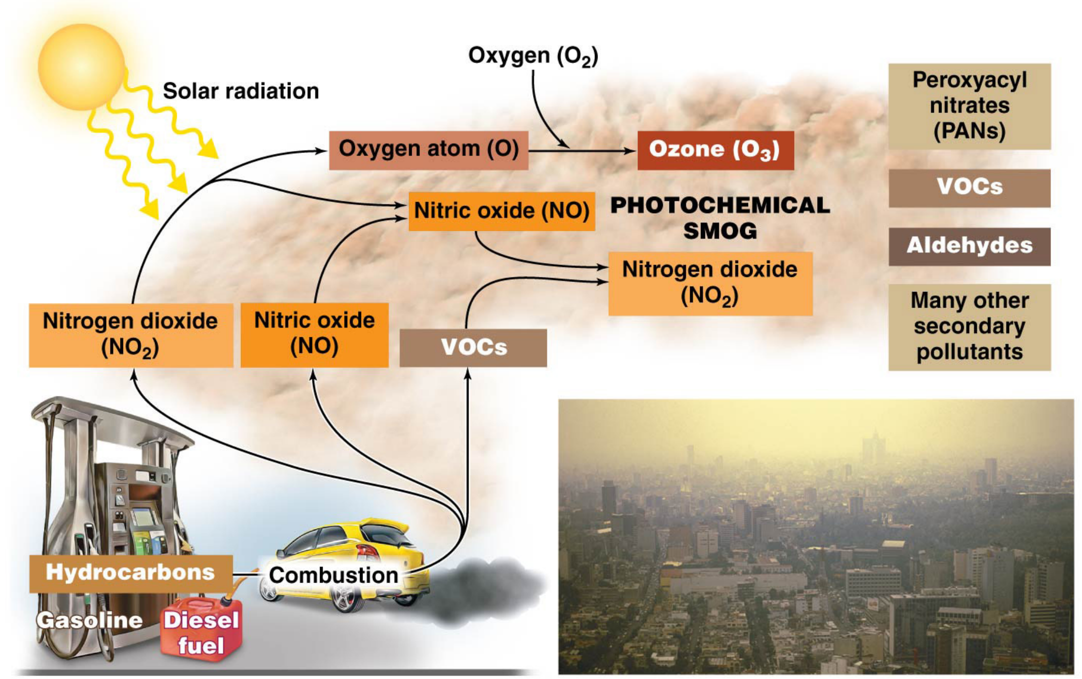
Photochemical smog (LA smog)

-appears as a brownish haze

- results when pollutants from automobile exhaust react amid exposure to light

- high levels of ground level ozone, nitrogen oxides, PM and a mix of other chemicals

- respiratory problems and other health issues



(a) Formation of photochemical smog (b) Photochemical smog over Mexico City

Let's Talk

Groups: 3-4 people (Include all names)

How do our daily choices impact air pollution in our communities and globally?

Think of your transportation choices, energy use/sources, products we buy